

TIC 370553309

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 11.540028 d
Epoch (t0) 1576.2791 BTJD
Duration 0.3500 d
Depth 2.621 % (26,206 ppm)
SNR 355.4

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

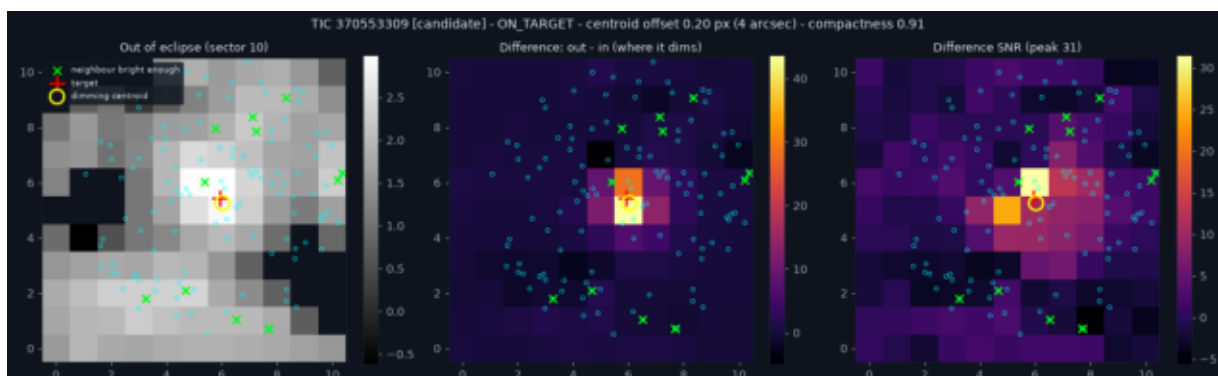
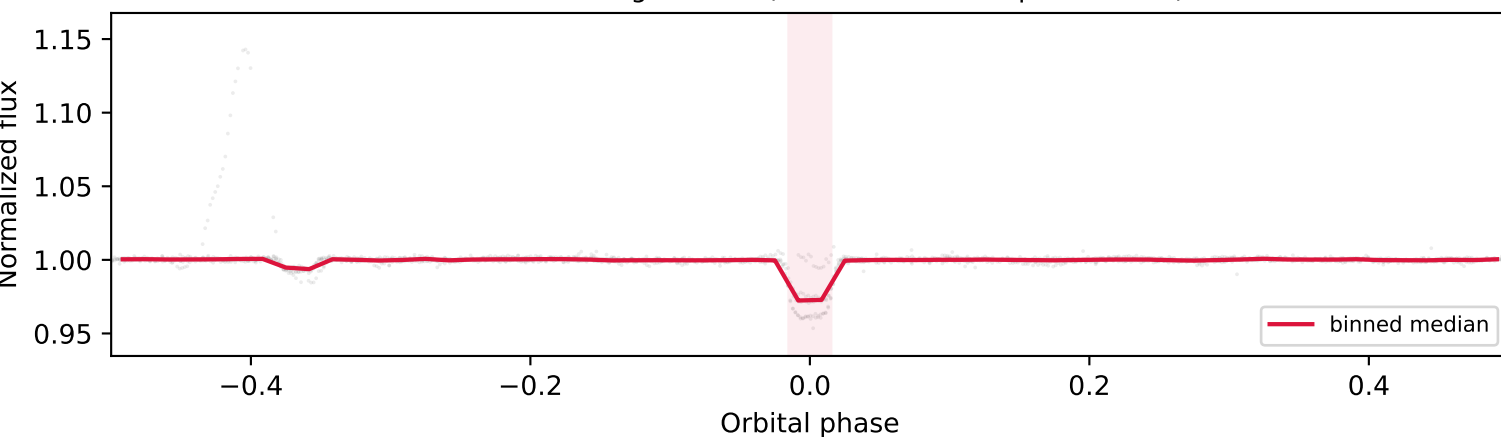
Eclipses observed 5 (gate ≥ 3)
Robust depth ratio 1.06 (gate ≥ 0.50)
Robust depth sigma 179.0 (gate ≥ 7.0)
Odd/even z-score 1.97 (consistent)
Secondary eclipse detected, phase -0.36, 5.7 sigma

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 10 light-curve dip in this sector: 126.6 sigma t0 refined by 0.0025 phase
Difference-image peak SNR 31.48 (gate ≥ 5.0) Compactness 0.91 (gate ≥ 0.50)
Dimming centroid offset 0.20 px (4 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 370553325
1.01 px away, dTmag +0.31 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



No caveats - clean detection on every check.